Results: Language, Topics, and Embeddings

RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, effects, patients, results, sleep, used, use, drug, studies, clinical, mg, using, placebo, cognitive, associated, effect, however, disorder.

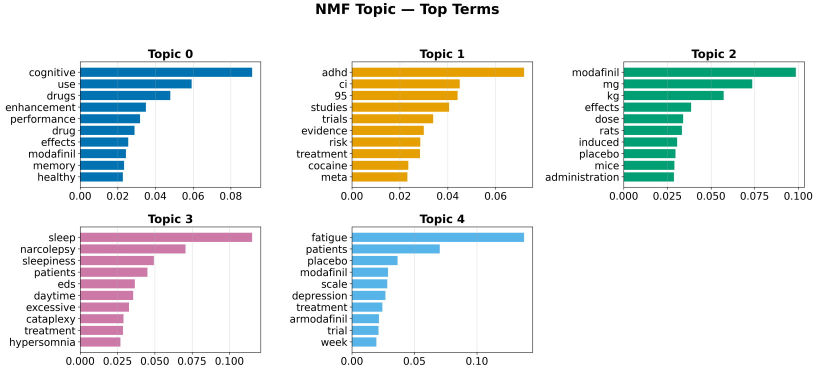
Table 3. NMF topics extracted from the corpus.

Topic	Top terms
0	cognitive, use, drugs, enhancement, performance, drug, effects, modafinil
1	adhd, ci, 95, studies, trials, evidence, risk, treatment
2	modafinil, mg, kg, effects, dose, rats, induced, placebo
3	sleep, narcolepsy, sleepiness, patients, eds, daytime, excessive, cataplexy
4	fatigue, patients, placebo, modafinil, scale, depression, treatment, armodafinil

RQ3: Topical and Linguistic Structure II

The topics reveal the thematic structure of the literature: Topic 0 centres on cognitive enhancement and neuroenhancement; Topic 1 addresses ADHD treatment and clinical evidence; Topic 2 covers pharmacological dose-response studies (including animal models); Topic 3 focuses on sleep disorders (narcolepsy, excessive daytime sleepiness); and Topic 4 addresses fatigue in psychiatric populations. These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

RQ3: Topical and Linguistic Structure III

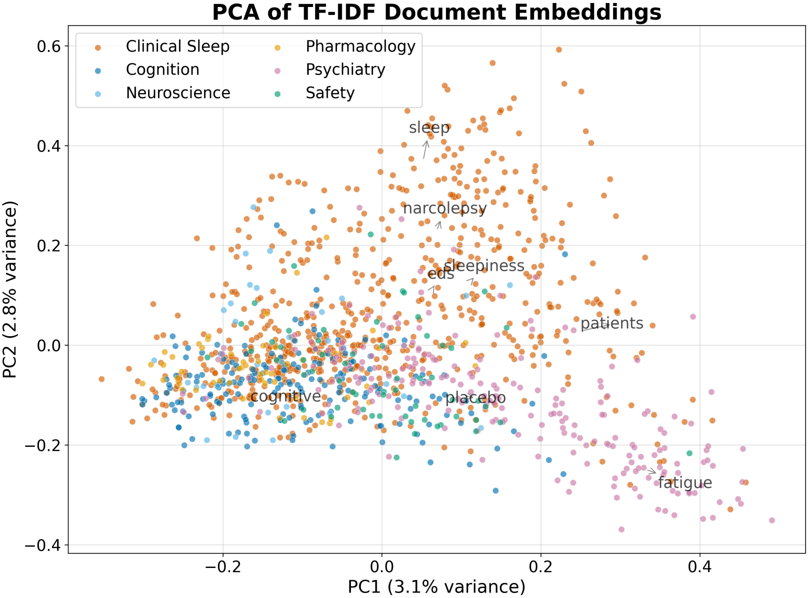


{{#fig

Document Embeddings I

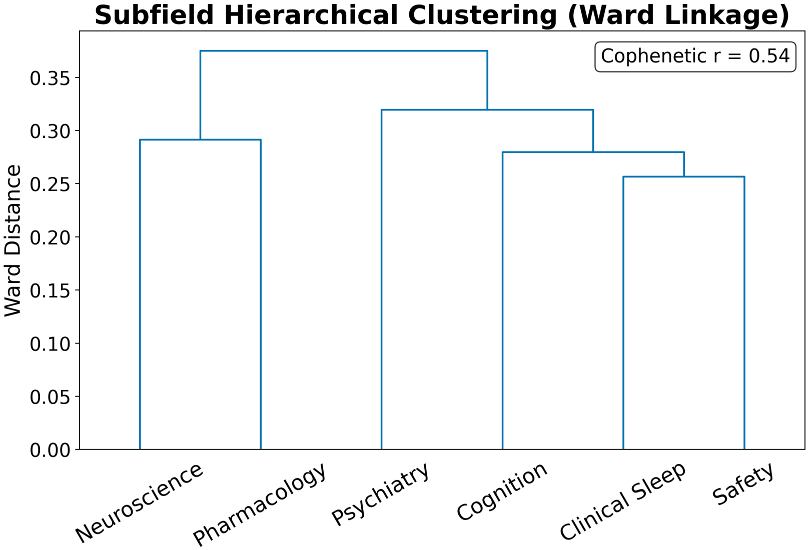
Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

Document Embeddings II



{{#fi

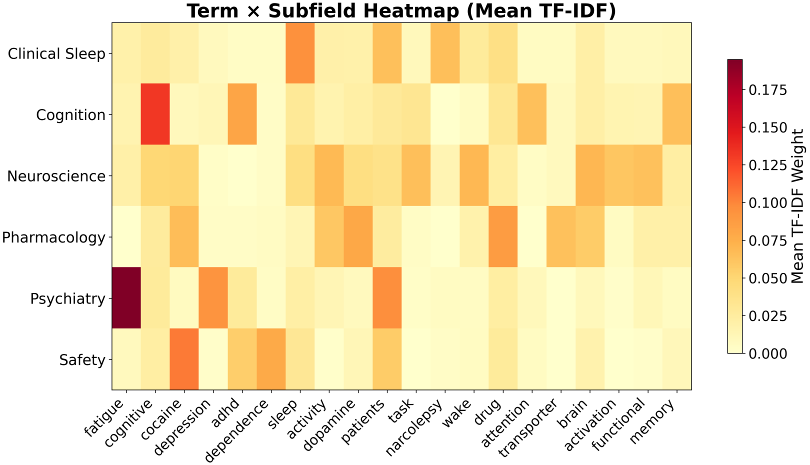
Document Embeddings III



{{#fig

Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.



{{#fig

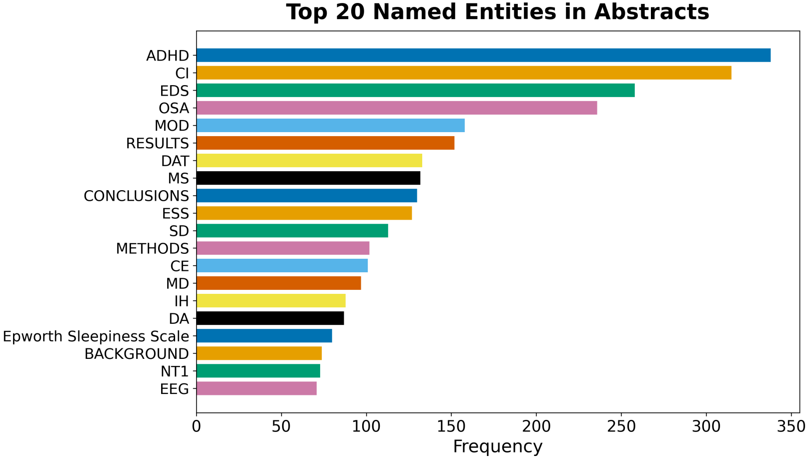
Named Entity Analysis I

Named entity extraction over the 1277 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

Table 4. Top named entities in abstracts.

Entity	Frequency
ADHD	338
CI	315
EDS	258
OSA	236
MOD	158
RESULTS	152
DAT	133
MS	132
CONCLUSIONS	130
ESS	127
SD	113
METHODS	102
CE	101
MD	97
IH	88

Named Entity Analysis II



{{#fig

Table 5. Top keyphrases by TF-IDF score.

Keyphrase	Score
available	0.3333
abstract	0.3333

Named Entity Analysis III

abstract available	0.3333
content	0.1053
access	0.1053
md	0.0870
jama	0.0826
cleveland	0.0763
modafinil	0.0741
depression	0.0741
substance	0.0694
drug	0.0667
conditions	0.0667
continuous	0.0667
continuous flow	0.0667

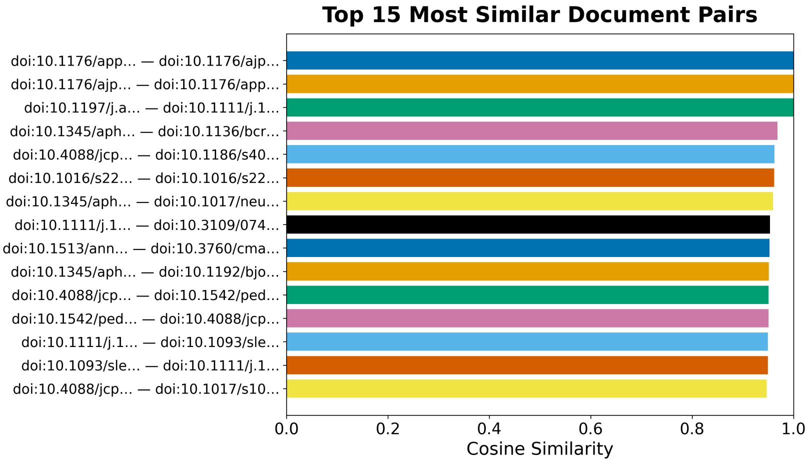
Embedding Similarity and Clustering I

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Paper A	Paper B	Similarity
doi:10.1176/appi.ajp.163.12.21	doi:10.1176/ajp.2006.163.12.21	1.0000
doi:10.1176/ajp.2006.163.12.21	doi:10.1176/appi.ajp.163.12.21	1.0000
doi:10.1197/j.aem.2005.08.013	doi:10.1111/j.1553-2712.2006.t	1.0000
doi:10.1345/aph.1h302	doi:10.1136/bcr.08.2011.4652	0.9687
doi:10.4088/jcp.09m05900gry	doi:10.1186/s40345-015-0034-0	0.9628
doi:10.1016/s2215-0366(18)3026	doi:10.1016/s2215-0366(25)0006	0.9621
doi:10.1345/aph.1h302	doi:10.1017/neu.2023.6	0.9598
doi:10.1111/j.1365-2869.2008.0	doi:10.3109/07420528.2011.6352	0.9538
doi:10.1513/annalsats.202006-6	doi:10.3760/cma.j.cn112147-202	0.9532
doi:10.1345/aph.1h302	doi:10.1192/bjo.2024.75	0.9517

Embedding Similarity and Clustering II



{{#fig

Embedding Similarity and Clustering III

Literature Corpus — Term Cloud

{{#fig

Embedding Similarity and Clustering IV